# Ahmed Sallam

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## PROFILE

Data Engineer experienced in building ETL/ELT data pipelines using Python and SQL within cloud and on-premises environments. Skilled in data warehousing with Snowflake, big data processing with Apache Spark & Hadoop, and workflow orchestration using Apache Airflow, with a focus on scalable and reliable data solutions for analytics and reporting.

## PROJECTS

- **Real-Time Fraud Detection Data Platform (Repo)**
    - Designed and implemented a production-style **real-time** data platform for financial transaction analytics and **fraud detection**.
    - Built a **Change Data Capture** (CDC) pipeline to ingest transactional data from MSSQL using Debezium and **Kafka** event streaming.
    - Developed **real-time processing** workflows using Apache Flink to **enrich events**, **engineer features**, and apply **ML-based** fraud scoring.
    - Orchestrated batch and streaming **ELT** workflows using Apache Airflow to load partitioned data into Snowflake.
    - Implemented **dimensional data models** and **transformations** using dbt following **Kimball methodology**.

    **Technologies:** Python, SQL, Kafka, Debezium, Apache Flink, Airflow, dbt, Snowflake, Docker, S3

- **EGX Big Data Pipeline (Repo)**
    - Built a production-grade **Lambda Architecture** pipeline for Egyptian Exchange (EGX) stock market data, processing daily OHLCV **batches and real-time** tick streams via **Kafka**.
    - Implemented **medallion data lake on HDFS** (Raw → Staging → Curated) with **PySpark** ETL computing technical indicators (SMA, EMA, RSI, MACD, Bollinger Bands).
    - Orchestrated 4 **Airflow DAGs** (batch ingestion, Spark ETL, Hive view refresh, streaming) served live data to **Power BI** via Simba Hive ODBC.

    **Technologies**: Python, PySpark, Apache Kafka, Hadoop HDFS, Hive, Airflow, Docker, yfinance, Power BI

- **GCP E-Commerce Data Platform (Repo)**
    - Built a scalable **end-to-end** data pipeline on **Google Cloud Platform** using serverless Python and Cloud Run to **ingest, process, and store** e-commerce data from **multiple sources** in BigQuery.
    - Designed **robust streaming data processing** with **deduplication** (exact + fuzzy matching), **error handling** via **dead-letter queues, and incremental data ingestion** to ensure data quality and reliability.
    - Implemented a **modern data architecture** combining data lake (GCS), data warehouse (BigQuery), and **real-time API layer** (FastAPI + Firestore caching) for efficient analytics and fast data access.
    - Developed **advanced analytics features** to generate actionable business insights.

    **Technologies:** Python, Google Cloud Platform (Cloud Run, Pub/Sub, Dataflow, BigQuery, Cloud Storage, Firestore, Cloud Scheduler), SQL

- **E-Commerce Data Warehouse (Repo)**
  - Designed and implemented an **end-to-end dimensional data warehouse** for retail analytics using **star schema** modeling.
  - Built **ETL pipelines** using SQL Server Integration Services (SSIS) to extract, transform, and load data from operational systems.
  - Implemented **Slowly Changing Dimensions** (SCD Type 2) for historical tracking and analytical accuracy.

  **Technologies:** SQL Server, T-SQL, SSIS, Data Modeling, ETL, Star Schema

## WORK EXPERIENCE

- **LLM Developer | G2I** **(Sep 2024 – Aug 2025) | Remote**
  - Supported the training and improvement of AI models by refining responses, rewriting Python code, and optimizing prompt structure.
  - Evaluated large language model (LLM) outputs for accuracy, clarity, and relevance across diverse use cases and edge cases.

## PUBLICATION & RESEARCH

- **Human Gait Recognition for Security Systems" - International Technology and Artificial Intelligence Forum (ITAF), Canadian International College (Oct 2024)** **View Publication**
  - Developed a novel security system utilizing computer vision and AI to analyze human gait patterns for non-intrusive identification.
  - Collaborated with a team of 7 researchers to achieve high accuracy rates in human identification through walking pattern analysis.

## TECHNICAL SKILLS

- **Languages:** Python, SQL
- **Methodologies:** ETL/ELT, Data Warehousing, Data Modeling, CDC, Batch & Streaming Pipelines
- **Data Platforms:** Snowflake, SQL Server
- **Orchestration:** Apache Airflow
- **Streaming:** Kafka, Flink, Spark
- **Big Data**: Apache Spark (PySpark), Apache Hadoop (HDFS, YARN), Hive
- **Cloud:** Google Cloud Platform (BigQuery, Dataflow, Pub/Sub, Cloud Run, Cloud Storage), Amazon S3, IAM Roles & Policies
- **Transformation:** dbt
- **Tools:** Git, Docker, Linux

## EDUCATION

- **B.Sc. in Artificial Intelligence** **Sep 2020 - Jul 2024**
- **Intelligent Security System (Graduation Project)** **Grade | A+**
  - Designed and implemented an AI-powered security system Integrated into a cohesive desktop application:
    - Gait recognition
    - Aggressive behavior detection
    - Car license plate recognition
    - Face recognition in crowded environments
    - Fall detection